LockwoodSTEM

Aerospace Engineering Design

Unit 2: CAD Modeling, Parametric Design & 3D Printing

Student learning deck • LockwoodSTEM aerospace-aligned curriculum

Skills

Fusion setup, sketches, constraints, features, parametric edits, and slicer prep.

Evidence

CAD screenshots, exported files, slicer settings, test prints, and revision notes.

Challenge

Design and justify a 3D printed tolerance tester for rocket assembly.

Model • Print • Test • Revise

Unit 2 Learning Arc

CAD turns documented ideas into accurate, editable, printable models.

Start with evidence

Use sketches and measurements as inputs.

Build controlled models

Use planes, constraints, dimensions, and features.

Prepare for printing

Check the CAD model and slicer setup.

Use the print as data

Test fit and recommend clearances for Unit 3.

Unit 2 Deliverables Map

Track these artifacts as the unit develops.

CAD evidence

Fusion project setup
Constrained and dimensioned sketches
Extrudes, cuts, fillets, revolves, and patterns
Screenshots showing revisions

Print evidence

STL/exported files
Slicer preview and settings
Tolerance tester print
Fit test recommendation

Your deliverables should prove how your design decisions were made.

Unit 2 Project: Tolerance Tester

Your final print is a test tool, not just a decoration.

Purpose

Measure and compare fit between printed features and rocket assembly parts.

Evidence

Use repeated holes, pegs, slots, or couplers to identify reliable clearances.

Recommendation

Use test results to recommend a clearance for Unit 3 rocket assembly.

Lesson 2.1

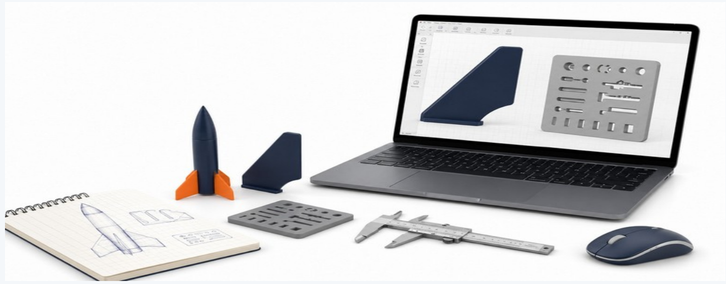
From Engineering Documentation to CAD

How do engineers turn sketches and measurements into digital models?

Lesson 2.1

From Engineering Documentation to CAD

Connect today's work to your Unit 2 CAD and 3D printing evidence.



From Engineering Documentation to CAD

Big ideas you should understand before you start the work

Big Ideas

1. CAD models should be based on documented measurements, not guesses.
2. Unit 1 sketches become a modeling plan.
3. Good planning prevents wrong-size digital parts.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

From Engineering Documentation to CAD

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Use your sketch as the source, not your memory.

Identify which dimensions control the shape.

Name files so they can be found later.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

From Engineering Documentation to CAD

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Notebook entry connects sketch evidence to a planned CAD model.

CAD file or screenshot is saved with a clear name.

Model decisions are explained with measurements.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.2

Fusion 360 Interface and File Setup

How do engineers set up a CAD workspace so work is organized and accurate?

Lesson 2.2

Fusion 360 Interface and File Setup

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Fusion 360 Interface and File Setup

Big ideas you should understand before you start the work

Big Ideas

1. The interface is a tool system: browser, timeline, sketch tools, view cube, and units.
2. File setup affects every model made afterward.
3. A clean project folder prevents lost or duplicated work.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Fusion 360 Interface and File Setup

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Create/open the correct project location.

Set units before modeling.

Save early with a professional file name.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Fusion 360 Interface and File Setup

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Fusion project is created or located.

Test file is saved correctly.

Student can identify key interface areas.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.3

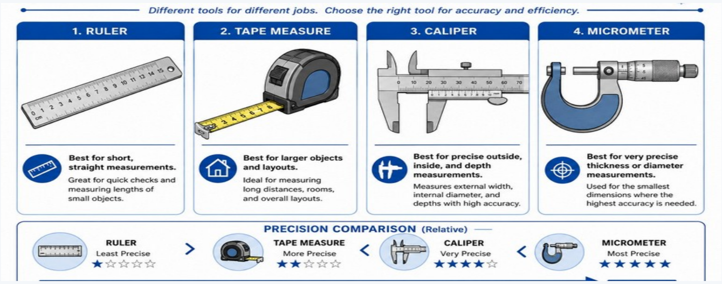
Sketch Planes and 2D Sketching

Why does every CAD model start with a carefully planned 2D sketch?

Lesson 2.3

Sketch Planes and 2D Sketching

Connect today's work to your Unit 2 CAD and 3D printing evidence.



Sketch Planes and 2D Sketching

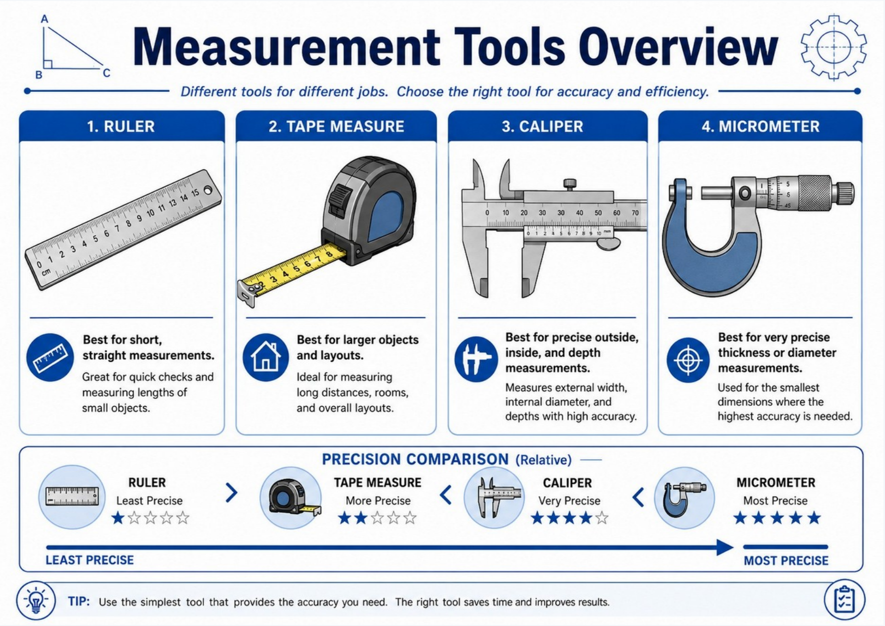
Big ideas you should understand before you start the work

Big Ideas

1 The sketch plane controls where geometry begins.

2 2D sketches define the base shape of most 3D parts.

3 Poor sketch placement creates confusing models.



Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Sketch Planes and 2D Sketching

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Choose a plane that matches the intended view.

Start with simple geometry before details.

Keep sketches readable and organized.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Sketch Planes and 2D Sketching

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Sketch plane is selected intentionally.

Base sketch is visible and understandable.

Student explains why the plane was chosen.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.4

Sketch Constraints

How do engineers control how a CAD sketch behaves when dimensions change?

Lesson 2.4

Sketch Constraints

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Sketch Constraints

Big ideas you should understand before you start the work

Big Ideas

1. Constraints describe relationships between sketch entities.
2. A fully defined sketch behaves predictably.
3. Uncontrolled sketches can shift unexpectedly.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Sketch Constraints

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Apply horizontal, vertical, parallel, perpendicular, tangent, and coincident constraints.

Watch how constraints change degrees of freedom.

Test the sketch by changing a dimension.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Sketch Constraints

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Constrained sketch is saved.

At least three constraint types are identifi[redacted].

Student explains why constraints improve model quality.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Sketch Constraints

Optional video resource to reinforce the lesson ideas.

What to watch for

Fusion 360: Sketch Constraints

how constraints control sketch behavior

why fully defined sketches are more predictable

what happens when a dimension changes

Scan or open

Open Video

https://www.youtube.com/watch?v=J_2If5zVp84

Lesson 2.5

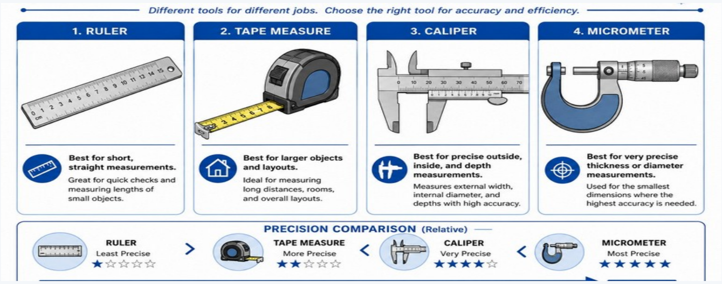
Dimensioned CAD Sketches

How do dimensions turn a rough CAD sketch into an accurate engineering model?

Lesson 2.5

Dimensioned CAD Sketches

Connect today's work to your Unit 2 CAD and 3D printing evidence.



Dimensioned CAD Sketches

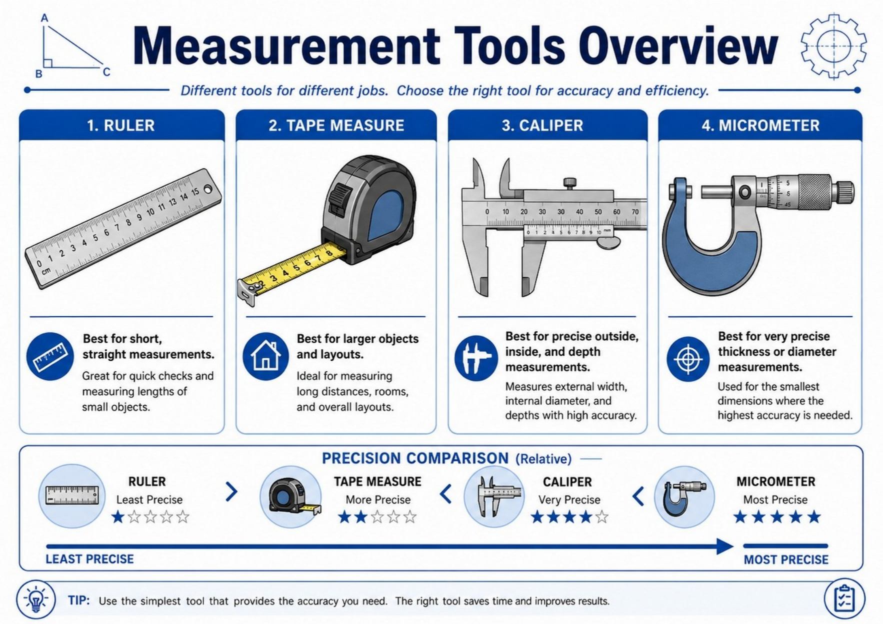
Big ideas you should understand before you start the work

Big Ideas

1 Dimensions convert rough geometry into measurable engineering intent.

2 Key dimensions should come from sketches, requirements, or measured objects.

3 Over-dimensioning can make sketches hard to edit.



Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Dimensioned CAD Sketches

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Add dimensions to control size and spacing.

Use consistent units.

Revise dimensions when the design intent changes.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Dimensioned CAD Sketches

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Sketch includes necessary dimensions.

Dimensions match documentation or requirements.

Student can identify controlling dimensions.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.6

Extrude Basics

How does a 2D sketch become a 3D CAD model?

Lesson 2.6

Extrude Basics

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Extrude Basics

Big ideas you should understand before you start the work

Big Ideas

1. Extrude adds depth to a closed profile.
2. Extrude distance controls thickness, length, or height.
3. Extrude direction matters for assembly and print planning.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Extrude Basics

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Select a closed profile.

Choose join, new body, or cut intentionally.

Check the 3D view after the feature is created.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Extrude Basics

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Basic extruded part is complete.

Feature is named or easy to identify.

Student can explain sketch-to-solid workflow.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.7

Cut Features and Holes

How do engineers remove material from a CAD model to create functional features?

Lesson 2.7

Cut Features and Holes

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Cut Features and Holes

Big ideas you should understand before you start the work

Big Ideas

1 Cuts create holes, slots, pockets, and clearance features.

2 Functional openings must be placed and dimensioned accurately.

3 Cut direction and depth affect whether the feature works.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Cut Features and Holes

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Sketch the removal area on the correct face.

Use cut distance or through-all when appropriate.

Inspect the model from multiple views.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Cut Features and Holes

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Part includes at least one accurate cut feature.

Holes or slots are dimensioned.

Student checks that the feature goes where intended.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.8

Fillets, Chamfers, and Design Intent

Why do engineers modify sharp edges in a CAD model?

Lesson 2.8

Fillets, Chamfers, and Design Intent

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Fillets, Chamfers, and Design Intent

Big ideas you should understand before you start the work

Big Ideas

1. Fillets and chamfers can improve safety, strength, fit, and printability.
2. Edge treatment should support the function, not just appearance.
3. Design intent explains why a feature exists.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Fillets, Chamfers, and Design Intent

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Identify edges that need treatment.

Apply fillets or chamfers with reasonable sizes.

Avoid changing critical fit surfaces without a reason.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Fillets, Chamfers, and Design Intent

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Part includes intentional edge treatment.

Student explains why each treatment was added.

Critical dimensions are still protected.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.9

Revolve Features: Nose Cone and Nozzle

How can engineers create round 3D parts from a 2D sketch?

Lesson 2.9

Revolve Features: Nose Cone and Nozzle

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Revolve Features: Nose Cone and Nozzle

Big ideas you should understand before you start the work

Big Ideas

1. Revolve creates circular parts from a profile and axis.
2. Profiles must be planned around the centerline.
3. Nose cones and nozzles are natural revolve examples.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Revolve Features: Nose Cone and Nozzle

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Sketch half of the profile.

Add a centerline or axis.

Use revolve angle to create the 3D body.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Revolve Features: Nose Cone and Nozzle

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Revolved part is complete.

Axis and profile are visible or explainable.

Student connects the feature to a rocket component.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Revolve Features: Nose Cone and Nozzle

Optional video resource to reinforce the lesson ideas.

What to watch for

Using the REVOLVE TOOL in Autodesk Fusion 360

how a 2D profile turns into a round part

where the revolve axis is selected

why nose cones and nozzles are good revolve examples

Scan or open

Open Video

https://www.youtube.com/watch?v=SY1fGI7XAz4

Lesson 2.10

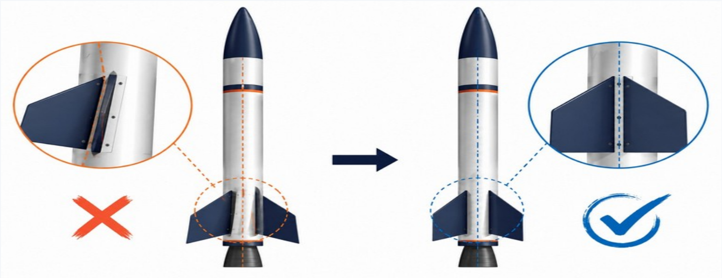
Patterns and Symmetry: Rocket Fins

How do engineers create repeated parts that are evenly spaced and aligned?

Lesson 2.10

Patterns and Symmetry: Rocket Fins

Connect today's work to your Unit 2 CAD and 3D printing evidence.



Patterns and Symmetry: Rocket Fins

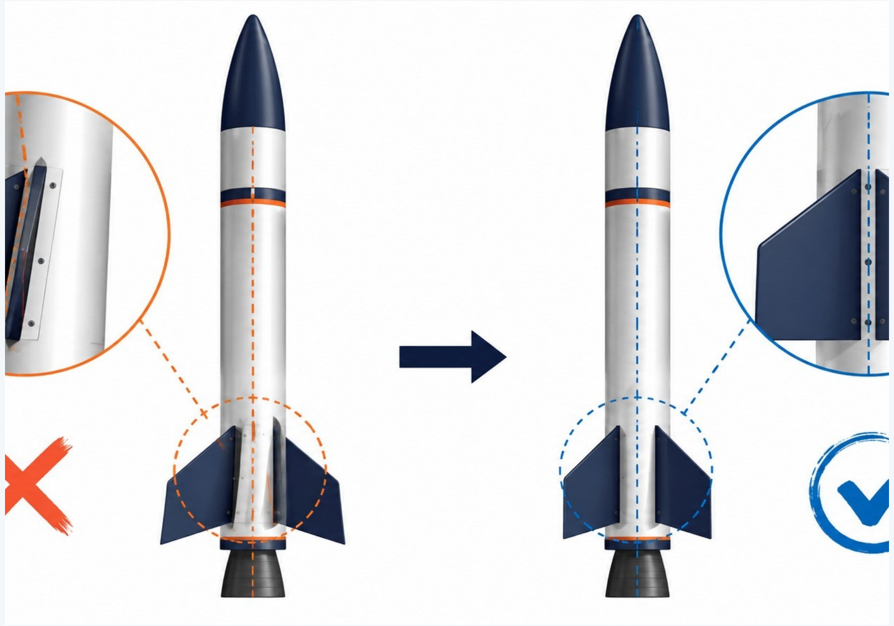
Big ideas you should understand before you start the work

Big Ideas

1 Patterns create consistent repeated features.

2 Symmetry improves alignment and reduces modeling errors.

3 Rocket fins must be evenly placed for balanced design.



Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Patterns and Symmetry: Rocket Fins

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Create or place one accurate fin feature first.

Use circular pattern when spacing around a centerline.

Check fin count, angle, and alignment.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Patterns and Symmetry: Rocket Fins

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Fin pattern or symmetry model is complete.

Student explains the pattern method used.

Alignment is checked visually and with CAD tools.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.11

Parametric Rocket Part Modeling

How can engineers design CAD models so they are easy to change later?

Lesson 2.11

Parametric Rocket Part Modeling

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Parametric Rocket Part Modeling

Big ideas you should understand before you start the work

Big Ideas

1 Parameters allow key values to update the model.

2 A parametric model is easier to revise than a fixed model.

3 Useful parameters connect to real design requirements.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Parametric Rocket Part Modeling

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Identify values that may change later.

Use named dimensions or editable parameters.

Test by changing one value and observing the result.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Parametric Rocket Part Modeling

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Model includes editable dimensions or parameters.

Student identifies which values control the design.

A quick revision test is documented.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.12

CAD Model Check and Revision

How do engineers check whether a CAD model is correct before moving forward?

Lesson 2.12

CAD Model Check and Revision

Connect today's work to your Unit 2 CAD and 3D printing evidence.

CAD Model Check and Revision

Big ideas you should understand before you start the work

Big Ideas

1 CAD review prevents printing the wrong part.

2 Models should be checked for size, units, features, and printability.

3 Revision is normal engineering work.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

CAD Model Check and Revision

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Inspect dimensions and units.

Compare the model to the design intent.

Revise the model before exporting.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

CAD Model Check and Revision

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

CAD check list is complete.

At least one revision is documented.

Final screenshot shows the corrected model.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.13

Preparing CAD Files for 3D Printing

How do engineers prepare a CAD model so it can be printed reliably?

Lesson 2.13

Preparing CAD Files for 3D Printing

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Preparing CAD Files for 3D Printing

Big ideas you should understand before you start the work

Big Ideas

1. A printable model must be exported correctly.
2. Orientation and geometry affect support, strength, and surface quality.
3. STL files should represent the final checked design.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Preparing CAD Files for 3D Printing

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Inspect the model before export.

Export the correct body or component.

Use clear file naming for print submission.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Preparing CAD Files for 3D Printing

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

STL/export file is prepared.

Print orientation is considered.

Student can explain why the file is ready.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.14

Slicer Basics and Print Settings

How do slicer settings affect print success, time, material, and accuracy?

Lesson 2.14

Slicer Basics and Print Settings

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Slicer Basics and Print Settings

Big ideas you should understand before you start the work

Big Ideas

1 A slicer converts a model into machine instructions.

2 Layer height, infill, walls, and supports change print performance.

3 Preview mode helps catch problems before printing.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Slicer Basics and Print Settings

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Import and orient the model.

Check supports, estimated time, and material use.

Preview layers before submitting.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Slicer Basics and Print Settings

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Slicer screenshot is captured.

Settings are recorded or justifi[illegible].

Student explains one trade-off in the settings.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Slicer Basics and Print Settings

Optional video resource to reinforce the lesson ideas.

What to watch for

PrusaSlicer Beginner Tutorial: Learn the Basics

how a model is imported and oriented

where print settings affect time and material

how preview helps catch problems before printing

Scan or open

Open Video

https://www.youtube.com/watch?v=_kIqMPNQNSw

Lesson 2.15

FabLab 3D Printer Certification

What must students know before using the 3D printers safely and responsibly?

Lesson 2.15

FabLab 3D Printer Certification

Connect today's work to your Unit 2 CAD and 3D printing evidence.

FabLab 3D Printer Certification

Big ideas you should understand before you start the work

Big Ideas

1. Certification protects people, tools, and materials.
2. Safe printer use includes setup, monitoring, and cleanup.
3. Responsible use means respecting the queue and documenting prints.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

FabLab 3D Printer Certification

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Review printer safety expectations.

Identify what to do before, during, and after a print.

Check print request or certification requirements.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

FabLab 3D Printer Certification

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Certification evidence is complete.

Student can describe safe printer behavior.

Student understands print queue expectations.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.16

Final 3D Printed Tolerance Tester Workday

How can a printed test tool help make better rocket assembly decisions?

Lesson 2.16

Final 3D Printed Tolerance Tester Workday

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Final 3D Printed Tolerance Tester Workday

Big ideas you should understand before you start the work

Big Ideas

1. Tolerance testers help compare real fits, not assumptions.
2. A useful tester includes multiple sizes or clear labeled features.
3. The print should produce data for the next design decision.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Final 3D Printed Tolerance Tester Workday

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Finalize the CAD model.

Check labels, clearances, and print orientation.

Submit or prepare the final print file.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Final 3D Printed Tolerance Tester Workday

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Final tolerance tester file is ready or submitted.

Tester features are labeled or documented.

Student predicts what data the tester will provide.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

Lesson 2.17

3D Printing Post-Processing

Why is a 3D print not always finished when it comes off the printer?

Lesson 2.17

3D Printing Post-Processing

Connect today's work to your Unit 2 CAD and 3D printing evidence.

3D Printing Post-Processing

Big ideas you should understand before you start the work

Big Ideas

1 Supports, brim, stringing, and rough edges can affect the final part.

2 Post-processing can change dimensions and fit.

3 Cleanup should be documented as part of fabrication.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

3D Printing Post-Processing

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Remove supports carefully.

Clean only what should be cleaned.

Measure again after cleanup if fit matters.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

3D Printing Post-Processing

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Post-processing steps are documented.

Part is inspected for damage or dimensional change.

Student records any cleanup that affects fit.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

3D Printing Post-Processing

Optional video resource to reinforce the lesson ideas.

What to watch for

How to Remove 3D Print Support Marks

how supports affect the final part

why sanding or cleanup can change dimensions

what should be documented after cleanup

Scan or open

Open Video

https://www.youtube.com/watch?v=iwFKuCesQiA

Lesson 2.18

Final Tolerance Tester Review and Recommendation

What clearance should be recommended for the next rocket assembly project?

Lesson 2.18

Final Tolerance Tester Review and Recommendation

Connect today's work to your Unit 2 CAD and 3D printing evidence.

Final Tolerance Tester Review and Recommendation

Big ideas you should understand before you start the work

Big Ideas

1. A recommendation should be based on fit evidence.
2. Testing should identify too tight, too loose, and acceptable fits.
3. The final conclusion connects Unit 2 to the rocket assembly unit.

Ask yourself: How does this CAD decision improve accuracy, editability, or printability?

Final Tolerance Tester Review and Recommendation

Use the lesson skill to make a more accurate, editable, or printable model.

Modeling moves

Test each feature consistently.

Record observations and measurements.

Choose a recommended clearance with evidence.

Quality checks

Does the model match the intended measurements?

Can another student understand the feature?

Will the part be printable or testable?

Good CAD work is controlled enough to revise and clear enough to explain.

Final Tolerance Tester Review and Recommendation

Save the strongest artifact from today for your Unit 2 portfolio.

Deliverable checklist

Final recommendation is written.

Evidence table or photos support the claim.

Student explains how the result will guide Unit 3.

Portfolio evidence

Dated notebook entry

Screenshot or exported file

Short claim explaining what the evidence proves

Your final portfolio should show how the model improved across the unit.

LockwoodSTEM

Unit 2 Close

CAD evidence becomes fabrication evidence.

Your tolerance tester should help make a real design recommendation for the next rocket assembly project.

Model

A controlled CAD model with meaningful features.

Print

A test part prepared with documented settings.

Recommend

A fit decision supported by evidence.